

Supernova

Along with many other elements, zinc atoms are formed inside supernovae (exploding giant stars).



Hemimorphite

This mineral contains zinc carbonate.

Hemimorphite was discovered by James Smithson, the founder of the **Smithsonian Institution**.



Zincite

Zinc oxide crystals are generally colourless.

## Uses

First-aid tape



Medical tapes that contain zinc oxide stop wounds from getting infected with microbes.

The zinc coating on this steel bridge protects it from rust.



Akashi Kaikyo Bridge, Kobe, Japan



American penny

This zinc coin is coated with copper.



Rubber boots

This flexible rubber is made stronger by adding zinc oxide.

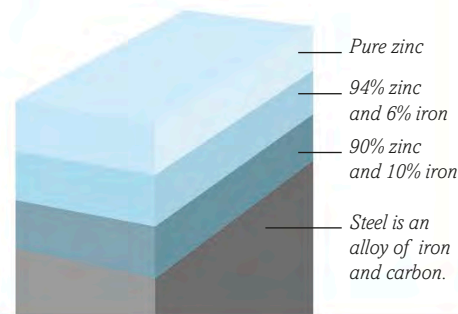
This soothing skin lotion contains a mixture of zinc compounds.

Calamine lotion



## GALVANIZED STEEL

Steel is protected from corrosion by coating it with zinc. Alloys of iron and zinc form in layers between the steel and pure zinc. This process is called galvanization.



range of applications. For example, a compound of zinc and oxygen called zinc oxide is used in **medical tape** and sunscreen. Zinc oxide can also be used to toughen the rubber used in **boots** and tyres. A compound of zinc and sulfur called zinc sulfide is used to make some

paints that glow in the dark. When pure zinc is exposed to air, the metal reacts with oxygen to form a protective layer of an oxide. This coating can prevent objects covered in zinc, such as **bridges**, from corroding easily.